Anthony Olevester

Robotics & AI Research Engineer

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🌐 Portfolio: https://anthony-olevester.github.io/Portfolio/
🔗 Research & Code: https://anthony-olevester.github.io

About-me

Hands-on Project Engineer with strong grounding in **robotics systems, AI/ML pipelines, and applied research engineering**. Experienced in building end-to-end intelligent systems that combine perception, learning, decision-making, and deployment. Comfortable working in lab-style environments, converting ideas into experiments, and delivering working prototypes aligned with funded research objectives.

Core Technical Skills

Robotics & Systems

- Robotic system design (perception → control → evaluation)
- Sensor data pipelines, monitoring & anomaly detection
- Simulation-driven development and testing

AI / Machine Learning

- Deep Learning (PyTorch, JAX)
- Multimodal AI (text, audio, image)
- Representation learning & embeddings
- Reinforcement / evolutionary methods (EvoJAX)

Data & Infrastructure

- Python-based research pipelines
- PostgreSQL, pgvector, FAISS, Milvus
- Experiment tracking, evaluation loops
- Dockerized reproducible environments

Robotics & Mechatronics Projects

Modular Vision-Guided Robotic Arm — Final Year Project

- Designed and built a **modular robotic arm** capable of mimicking human motion using vision recognition
- Implemented camera-based perception to translate observed movements into real-time actuator control
- Created a **modular mechanical architecture** allowing interchangeable arm segments and end-effectors
- Progressed from scrap-material prototypes to a **fully 3D-printed functional arm** through iterative design
- Demonstrated real-world motion mimicry and system extensibility in public video demonstrations

Smart Insulin Pen Step Counter — Biocon Internship

- Developed a **custom KiCad-designed PCB step counter** attachable to a standard insulin pen to convert it into a smart injector
- Tracked mechanical dose delivery steps and logged injection usage
- Enabled communication with a mobile device for data monitoring
- Engineered under tight constraints of size, power consumption, and reliability
- Resulted in internal recognition and extended technical engagement within the company

Motorized Intelligent Pen Injector — Advanced Medical Mechatronics Project

- Worked on a **motorized insulin injector system** designed to interpret patient glucose levels and compute appropriate dosage
- Implemented embedded control logic for automated, safe insulin delivery
- Focused on closed-loop decision-making and actuation
- Initiated early-stage intellectual property documentation prior to exit

Research & Project Experience

Rapture Twelve Private Limited — *Lead Research Engineer*

2022 – 2026

Led multiple applied research and engineering projects spanning AI systems, automation, and intelligent monitoring.

Relevant Research Contributions:

- Designed **AI-driven monitoring systems** using time-series data, anomaly detection, and statistical metrics (useful for robotics telemetry & control monitoring)
- Built **perception pipelines** combining OCR, speech-to-text, and vision models into structured datasets
- Implemented **iterative evaluation loops** for model validation and correction

- Developed modular, experiment-friendly codebases suitable for collaborative research teams

Model Training & Learning Systems — *Research Projects*

Evolutionary Logic Circuits (EvoJAX + Bitwise Circuit Core)

- Trained populations of logic-circuit genomes via mutation, crossover, and selection
- Target task: 4-bit × 4-bit multiplication (8-bit output)
- Objective-driven search over hypothesis space with fitness evaluation and generalization pressure
- Designed simulation environment and evaluation pipeline (evolutionary ML training, non-gradient)

Transformer-Guided Genome Mutation

- Trained a Transformer model to predict improved mutations for genome representations
- Dataset: (bad genome → improved genome) pairs generated from EvoJAX simulations
- Program-improvement learning (program synthesis / meta-learning adjacent)
- Compared transformer-guided mutations against random mutations in controlled experiments

RNA 3D Structure Ensemble Correction Model

- Trained a correction MLP to refine 3D structures using multi-model inputs
- Inputs: predicted coordinates, confidence scores, sequence-level embeddings
- Outputs: refined structures optimized via structural accuracy metrics (e.g., TM-score / RMSD)
- Implemented iterative refinement and validation loops

Publications / Research Work

Deterministic Pairing & Residual Correction for Sim-to-Sim Robotics Alignment

- Developed a governance-driven pipeline to align PyBullet and MuJoCo dynamics
- Implemented deterministic pairing, alignment gating, and long-horizon validation
- Demonstrated stable residual correction under contact-rich scenarios
- Focused on simulator consistency for control and planning reliability

Other Research

Curvature–Information Principle (CIP) — *Independent Research Project*

- Developed a computational framework linking operator dynamics and information geometry
- Implemented controlled simulation pipelines with measurable evaluation metrics

Selected Tools & Frameworks

- **Languages:** Python, JavaScript, SQL
- **ML:** PyTorch, JAX, HuggingFace
- **Robotics-adjacent:** Simulation pipelines, control evaluation loops
- **Vision & Audio:** WhisperX, PaddleOCR
- **Design & Simulation Tools:** Unity (3D simulation & visualization), SolidWorks (mechanical design), KiCad (PCB design)
- **Infra:** Docker, Linux, GitHub Actions

Education

B.Tech — National Institute of Technology, Puducherry
2018 – 2022

Relevant focus: Robotics projects, systems engineering, applied computation